

1 Problem 1

In this text the questions for problem 1 for assignment 2 is presented, solved and answered. the system that is being analyzed in this problem is the following:

$$\frac{\partial n}{\partial t} = rn \left(1 - \frac{n}{K}\right) - \frac{An}{1 + n/B} + \frac{\partial^2 n}{\partial x^2} \quad (1)$$

1.1 a)

For this first assignment the equation for the system is to be transformed to dimensionless parameters this is done by the following substitutions:

$$\tau = At \quad (2)$$

$$\xi = x \sqrt{\frac{A}{D}} \quad (3)$$

$$u(\xi, \tau) = \frac{n(x, t)}{B} \quad (4)$$

To begin the transformation we start with evaluate the following:

$$\frac{\partial u}{\partial \tau} = \frac{\partial u}{\partial t} \frac{\partial t}{\partial \tau} = \frac{1}{BA} \cdot \frac{\partial n}{\partial t} \quad (5)$$

$$\frac{\partial^2 u}{\partial \xi^2} = \frac{\partial^2 u}{\partial x^2} \left(\frac{\partial x}{\partial \xi}\right)^2 = \frac{D}{BA} \cdot \frac{\partial^2 n}{\partial x^2} \quad (6)$$

With these two equations we can transform our system:

$$\frac{\partial u}{\partial \tau} = \frac{1}{BA} \left(rBu \left(1 - \frac{Bu}{K}\right) - \frac{ABu}{1 + u} + AB \frac{\partial^2 u}{\partial \xi^2} \right) = \rho u \left(1 - \frac{u}{q}\right) - \frac{u}{1 + u} + \frac{\partial^2 u}{\partial \xi^2} \quad (7)$$

Where $\rho = \frac{r}{A}$ and $q = \frac{K}{B}$.

To get the steady states of the system we set equation 7 equal to zero and solve for u. In the question for this assignment the diffusion term is neglected because the spatially homogenous solutions is sought for.

$$u \left(\rho \left(1 - \frac{u}{q}\right) - \frac{1}{1 + u} \right) = 0 \implies u_1^* = 0 \quad (8)$$

If u isn't zero the equation simplifies to the following:

$$\rho \left(1 - \frac{u}{q}\right) - \frac{1}{1 + u} = 0 \quad (9)$$

solving this expression for u gives the following result:

$$u_{2,3}^* = \frac{1}{2} \left(q - 1 \pm \sqrt{(q - 1)^2 - 4q\left(\frac{1}{\rho} - 1\right)} \right) \quad (10)$$

which gives the approximate answers $u_2^* = 1.438$ and $u_3^* = 5.562$.

1.2 b)

In this section we have analyzed 3 different starting positions. The function for the starting condition was the following:

$$u(\epsilon, 0) = \frac{u_0}{1 + e^{\epsilon - \epsilon_0}} \quad (11)$$

where the value of u_0 decides how high the concentration will be from zero up to ϵ_0 and after a exponential decay goes down to zero. We will also try to find a traveling wave for every simulation and if one is found, the traveling velocity c will be calculated and the phase plot for a given time will be shown with respect to u and $v = \frac{\partial u}{\partial \xi}$. We will also calculate the stability of the stable points found in question a) given the numerically estimated value of c . This will be done by the taking the following system from equation 7:

$$\frac{\partial u}{\partial \xi} = v \quad (12)$$

$$\frac{\partial v}{\partial \xi} = \frac{u}{1+u} - \rho u \left(1 - \frac{u}{q}\right) + \frac{\partial u}{\partial \tau} \quad (13)$$

The transformation for a traveling wave to stand still is given by $z = \xi - ct$ where c is the wave velocity, and with this we can do the following substitutions:

$$\frac{\partial u}{\partial \tau} = \frac{\partial u}{\partial z} \frac{\partial z}{\partial \tau} = -c \frac{\partial u}{\partial z} \quad (14)$$

$$\frac{\partial u}{\partial \xi} = \frac{\partial u}{\partial z} \frac{\partial z}{\partial \xi} = \frac{\partial u}{\partial z} = v \quad (15)$$

This means that equation 13 becomes:

$$\frac{\partial v}{\partial \xi} = \frac{u}{1+u} - \rho u \left(1 - \frac{u}{q}\right) - cv \quad (16)$$

The jacobian of this system therefore becomes:

$$J = \begin{pmatrix} 0 & 1 \\ \frac{1}{(u+1)^2} - \rho \left(1 - \frac{u}{q}\right) & -c \end{pmatrix} \quad (17)$$

and solving the eigenvalues for this matrix generates the following equation for the eigenvalues:

$$\lambda_{1,2} = \frac{1}{2} \left(-c \pm \sqrt{c^2 + 4 \left(\frac{1}{(1+u)^2} - \rho \left(1 - \frac{2u}{q}\right) \right)} \right) \quad (18)$$

1.2.1 i)

In figure 1 we can see the the first simulation. We can see that quickly forms a traveling wave that goes to the right where the concentration is rising from u_1^* to u_3^* . In figure 2 we calculated the wave velocity to $c = 0.181$. Substitution with numerical values for $u = u_1^*$ and $u = u_3^*$ in equation 18, we get that the the eigenvalues are real-valued with a negative and a positive value. This therefore means that the classification of the stable points are both saddle-points.

$$\xi_0 = 20.00, u_0 = 5.56$$

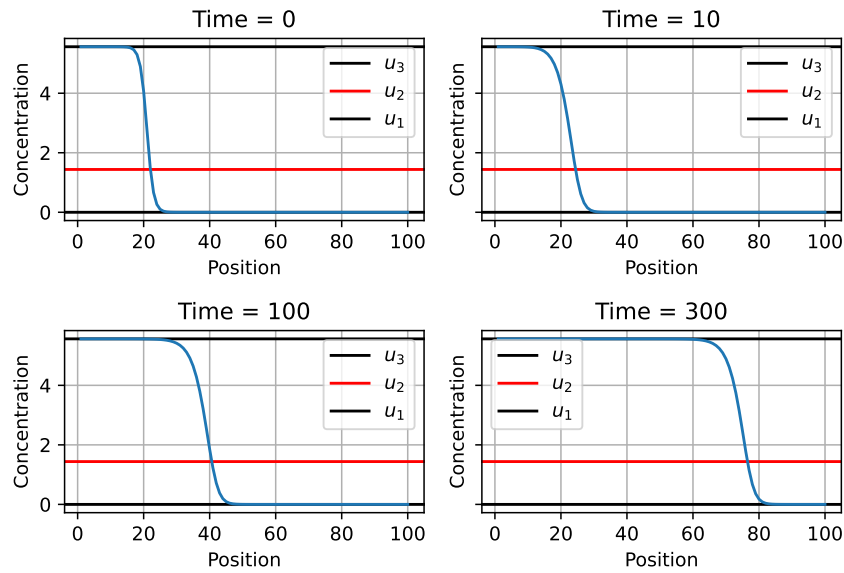


Figure 1: Figure over different time steps of the simulation with the first starting configuration.

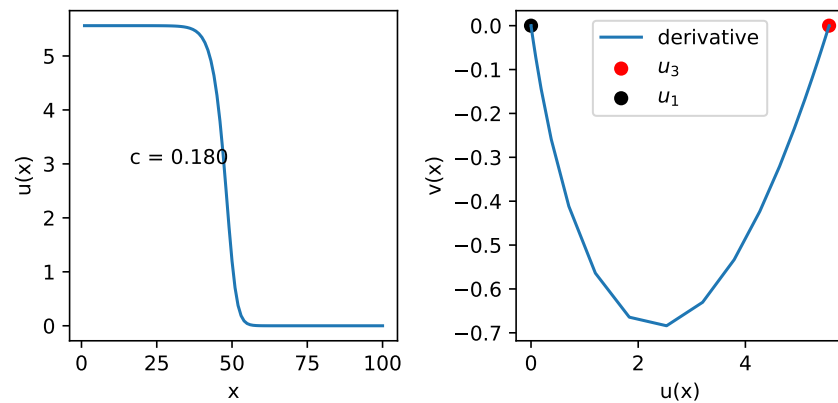


Figure 2: Figure over the trajectory of the traveling wave and the phase plot of the same trajectory.

1.2.2 ii)

In figure 3 we see different time step over the simulation done with the second starting condition. We see in the figure that it slowly decays from the unstable concentration u_2 to the stable concentration u_1 from the left. In figure 4 we calculated the final wave velocity to be -0.633 and paired with the values of the stable points it was calculated that the eigenvalues for u_1 was two real-values with opposite sign, which means a saddle-point and for the u_2 it was a complex number with both two positive real parts which mean that the stable point is a unstable spiral.

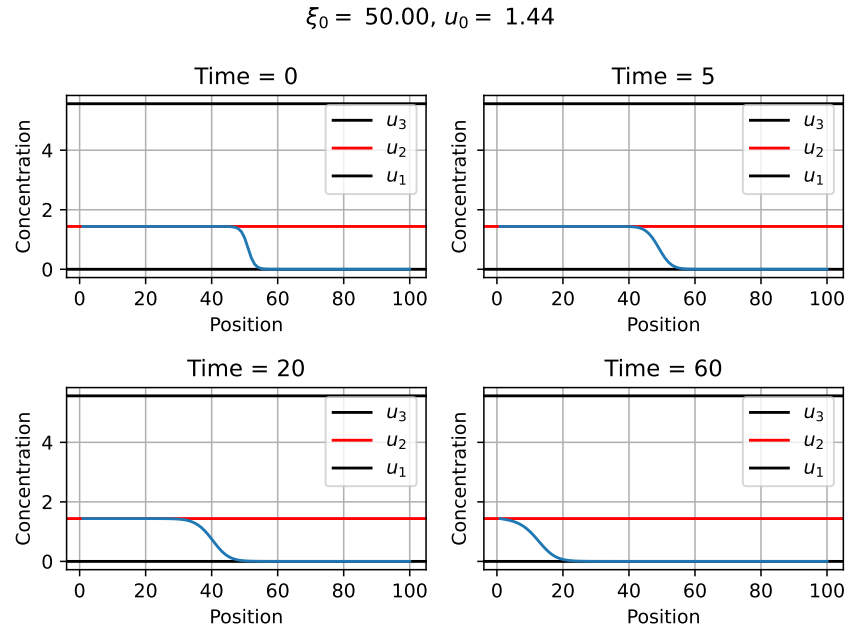


Figure 3: Figure over the simulation with the second starting configuration.

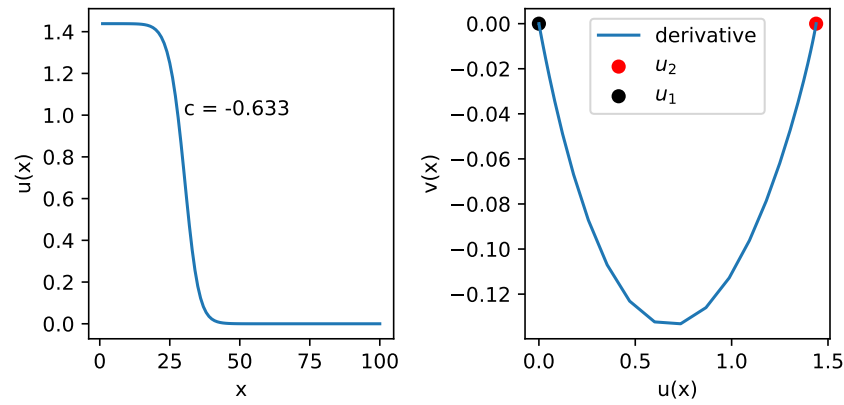


Figure 4: Figure over the trajectory of the traveling wave and the phase plot of the same trajectory.

1.2.3 iii)

In figure 5 we see the simulation of the third starting configuration. We see that between $t = 0$ and $t = 40$ the left side grows from u_2^* towards u_3^* and then creates a traveling wave after this, similar to the first starting configuration. And as we can see in figure 6 we get the same wave velocity $c = 0.181$ which therefore tells us that we get the same eigenvalues and therefore the same classification, two saddle-points.

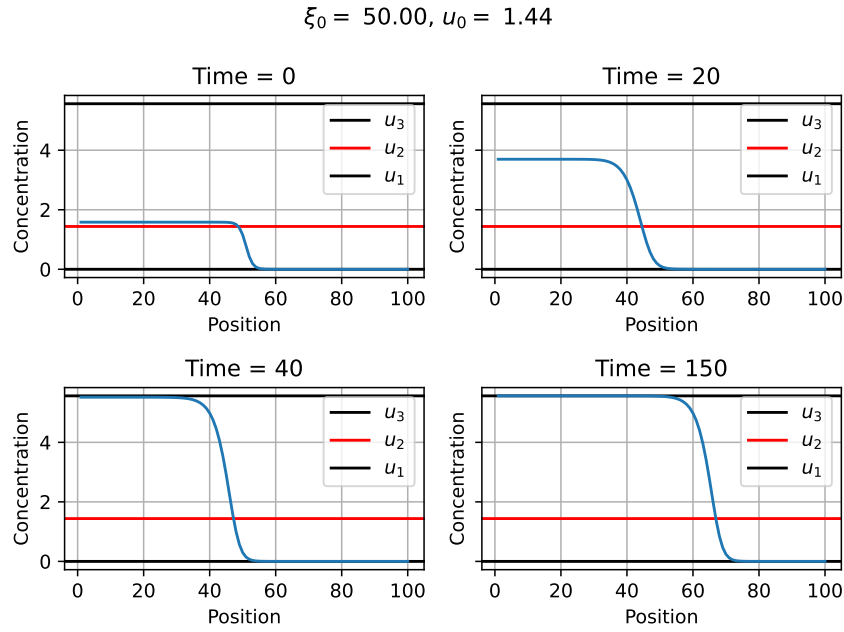


Figure 5: Figure over the simulation of the third starting configuration.

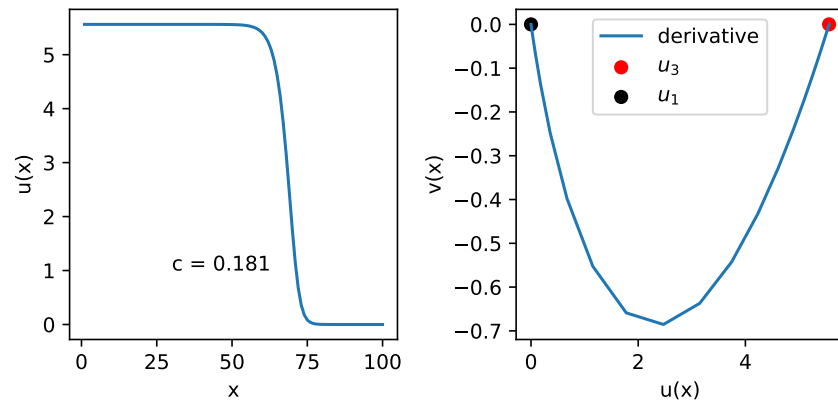


Figure 6: Figure over a trajectory of a traveling wave from the simulation and the phase plot over the same trajectory.

2 c)

In this section we switch the configuration to the following equation instead:

$$u(\xi, 0) = u_0 e^{-(\xi - \xi_0)^2} \quad (19)$$

This equation creates, as can be seen in figure 7 and 8 a much smaller starting configuration centered around ϵ_0 .

In figure 7 we see the simulation of the first modified starting configuration. We can clearly see that it quickly decays and dies down to u_1^* with no traveling wave being created.

$$\xi_0 = 50.00, u_0 = 5.56$$

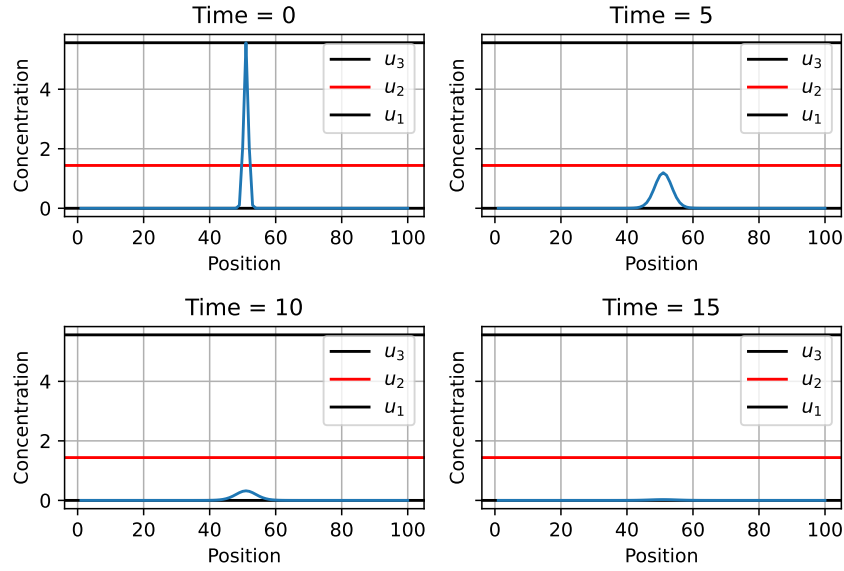


Figure 7: Figure over the first modified starting configuration.

In figure 8 it first starts to decrease, but as it spreads it never fully goes down to u_1^* but instead goes up again towards u_3^* . It still does not create a traveling wave because of it traveling at the same time in both directions and can therefore not be transformed to a static wave with the transformation $z = \xi - ct$.

$$\xi_0 = 50.00, u_0 = 16.68$$

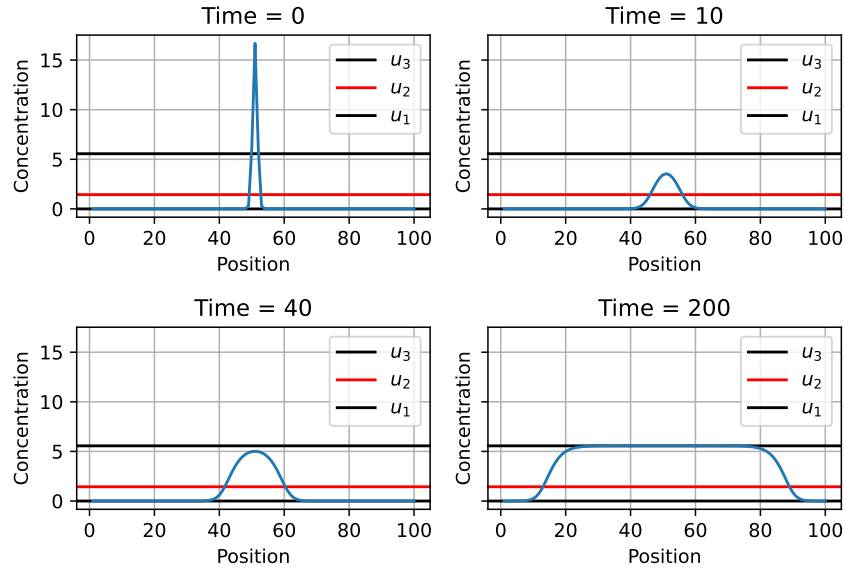


Figure 8: Figure over the second modified starting configuration.

3 code

```
import numpy as np
import matplotlib.pyplot as plt

def euler_first_derivative(f, h):
    first_derivative = (np.roll(f, -1) - np.roll(f, 1)) / 2 * h
    first_derivative = np.delete(first_derivative, [0, -1])
    return first_derivative

def euler_second_derivative(f, h):
    second_derivative = (np.roll(f, -1) + np.roll(f, 1) - 2 * f) / (h**2)
    return second_derivative

def system(u: np.ndarray, rho: float, q: float, d2udxi2: np.ndarray) -> np.ndarray:
    dudt = rho * u * (1 - u / q) - u / (1 + u) + d2udxi2
    return dudt

def start_concentration(u_0: float, xi_0: float, habitat_size: int) -> np.ndarray:
    index = np.arange(0, habitat_size, dtype=int)
    start_concentration = u_0 / (1 + np.exp(index - xi_0))
    return start_concentration

def start_concentration2(u_0: float, xi_0: float, habitat_size: int) -> np.ndarray:
    index = np.arange(0, habitat_size, dtype=int)
    start_concentration = u_0 * np.exp(-((index - xi_0) ** 2))
    return start_concentration

def run_iteration(
    n_i: np.ndarray, rho: float, q: float, time_step: float, habitat_step: float = 1.0
) -> np.ndarray:
    d2dxi2 = euler_second_derivative(n_i, habitat_step)
    d2dxi2[0] = 0.0
    d2dxi2[-1] = 0.0
    dudt = system(n_i, rho, q, d2dxi2)
    n_new = dudt * time_step + n_i
    return n_new

def run_simulation(
    u_0: float,
    xi_0: float,
    rho: float,
    q: float,
    habitat_size: int,
    sim_time: int,
    time_step: float,
    second_init=False,
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):
    sim_steps = int(sim_time / time_step)
    simulation_matrix = np.zeros((habitat_size, sim_steps))
    if second_init:
        simulation_matrix[:, 0] = start_concentration2(u_0, xi_0, habitat_size)
    else:
        simulation_matrix[:, 0] = start_concentration(u_0, xi_0, habitat_size)

    for idx in range(0, sim_steps - 1):
        simulation_matrix[:, idx + 1] = run_iteration(
            simulation_matrix[:, idx], rho, q, time_step
        )
    return simulation_matrix

def get_velocity_of_wave(
    simulation_matrix, u_stable, t_min, t_max, delta_step, time_step
) -> np.ndarray:
    index = np.arange(t_min, t_max, delta_step)
    c_array = np.zeros_like(index, dtype=float)

    conc_search = u_stable / 2
    for i, time in enumerate(index):
        x_min = np.abs(simulation_matrix[20:, time] - conc_search).argmin()
        x_max = np.abs(simulation_matrix[20:, time + delta_step] - conc_search).argmin()

        c_array[i] = ((x_max - x_min) / (delta_step)) / time_step
    return c_array

def plot1():
    rho = 0.5
    q = 8.0
    u_01 = 1 / 2 * (q - 1 + np.sqrt((q - 1) ** 2 - 4 * q * (1 / rho - 1)))
    u_02 = 1 / 2 * (q - 1 - np.sqrt((q - 1) ** 2 - 4 * q * (1 / rho - 1)))

    xi_0 = 20
    time_step = 0.1
    habitat_step = 1
    sim_time = 1000
    habitat_size = 100

    sim_matrix = run_simulation(u_01, xi_0, rho, q, habitat_size, sim_time, time_step)

    plot_steps = np.array([0, 100, 1000, 3000])
    x = np.arange(1, habitat_size + 1, 1)
    fig, ax = plt.subplots(2, 2, sharey=True)
    ax = ax.flatten()

    for idx, time in enumerate(plot_steps):
        ax[idx].axhline(u_01, label=r"$u_3$", c="black")
        ax[idx].axhline(u_02, label=r"$u_2$", c="red")
        ax[idx].axhline(0.0, label=r"$u_1$", c="black")
        ax[idx].plot(x, sim_matrix[:, time])

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    ax[idx].set_xlabel("Position")
    ax[idx].set_ylabel("Concentration")
    ax[idx].set_title(f"Time = {time*time_step:.0f}")
    ax[idx].grid()
    ax[idx].legend()

fig.suptitle(rf"$\xi_0 =\$ {xi_0:.2f}, $u_0 =\$ {u_01:.2f}")
fig.tight_layout()
fig.savefig("../report/figures/1b1.pdf")

t_min = int(50 / time_step)
t_max = int(250 / time_step)
delta_step = int(0.1 / time_step)
t_mid = int((t_max + t_min) / 2)

c = get_velocity_of_wave(sim_matrix, u_01, t_min, t_max, delta_step, time_step)

dudxi = euler_first_derivative(sim_matrix[:, t_mid], habitat_step)

fig, ax = plt.subplots(1, 2, figsize=(7, 3))
ax[0].plot(x, sim_matrix[:, t_mid])
ax[0].text(20, 5, f"c = {np.mean(c):.3f}")
ax[1].plot(sim_matrix[1:-1, t_mid], dudxi, label="derivative")
ax[1].scatter(sim_matrix[1, t_mid], dudxi[0], label="start point", color="red")
ax[1].scatter(sim_matrix[-2, t_mid], dudxi[-1], label="end point", color="black")
ax[1].legend()

ax[0].set_xlabel("x")
ax[0].set_ylabel("u(x)")
ax[1].set_xlabel("u(x)")
ax[1].set_ylabel("v(x)")

fig.tight_layout()
fig.savefig("../report/figures/1b1standi3gwave.pdf")

def plot2():
    rho = 0.5
    q = 8.0
    u_01 = 1 / 2 * (q - 1 + np.sqrt((q - 1) ** 2 - 4 * q * (1 / rho - 1)))
    u_02 = 1 / 2 * (q - 1 - np.sqrt((q - 1) ** 2 - 4 * q * (1 / rho - 1)))
    xi_0 = 50.0
    time_step = 0.1
    sim_time = 1000
    habitat_size = 100
    habitat_step = 1

    sim_matrix = run_simulation(u_02, xi_0, rho, q, habitat_size, sim_time, time_step)

    plot_times = np.array([0, 50, 200, 600])
    x = np.arange(1, habitat_size + 1, 1)
    fig, ax = plt.subplots(2, 2, sharey=True)
    ax = ax.flatten()

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for idx, time in enumerate(plot_times):
    ax[idx].axhline(u_01, label=r"$u_3$", c="black")
    ax[idx].axhline(u_02, label=r"$u_2$", c="red")
    ax[idx].axhline(0.0, label=r"$u_1$", c="black")
    ax[idx].plot(x, sim_matrix[:, time])
    ax[idx].set_xlabel("Position")
    ax[idx].set_ylabel("Concentration")
    ax[idx].set_title(f"Time = {time*time_step:.0f}")
    ax[idx].grid()
    ax[idx].legend()

fig.suptitle(rf"$\xi_0 = $ {xi_0:.2f}, $u_0 = $ {u_02:.2f}")
fig.tight_layout()
fig.savefig("../report/figures/1b2.pdf")

t_min = int(20 / time_step)
t_max = int(50 / time_step)
delta_step = int(0.1 / time_step)

t_mid = int((t_max + t_min) / 2)

c = get_velocity_of_wave(sim_matrix, u_02, t_min, t_max, delta_step, time_step)
dudxi = euler_first_derivative(sim_matrix[:, t_mid], habitat_step)

fig, ax = plt.subplots(1, 2, figsize=(10, 5))
ax[0].plot(x, sim_matrix[:, t_mid])
ax[0].text(30, 1, f"c = {np.mean(c):.3f}")
ax[1].plot(sim_matrix[1:-1, t_mid], dudxi, label="derivative")
ax[1].scatter(sim_matrix[1, t_mid], dudxi[0], label="start point", color="red")
ax[1].scatter(sim_matrix[-2, t_mid], dudxi[-1], label="end point", color="black")
ax[1].legend()

ax[0].set_xlabel("x")
ax[0].set_ylabel("u(x)")
ax[1].set_xlabel("u(x)")
ax[1].set_ylabel("v(x)")

fig.tight_layout()
fig.savefig("../report/figures/1b2standingwave.pdf")

def plot3():
    rho = 0.5
    q = 8.0
    u_01 = 1 / 2 * (q - 1 + np.sqrt((q - 1) ** 2 - 4 * q * (1 / rho - 1)))
    u_02 = 1 / 2 * (q - 1 - np.sqrt((q - 1) ** 2 - 4 * q * (1 / rho - 1)))
    xi_0 = 50.0
    time_step = 0.1
    sim_time = 1000
    habitat_size = 100
    habitat_step = 1

    sim_matrix = run_simulation(
        u_02 * 1.1, xi_0, rho, q, habitat_size, sim_time, time_step

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)

plot_times = np.array([0, 200, 400, 800])
x = np.arange(1, habitat_size + 1, 1)
fig, ax = plt.subplots(2, 2, sharey=True)
ax = ax.flatten()

for idx, time in enumerate(plot_times):
    ax[idx].axhline(u_01, label=r"$u_3$", c="black")
    ax[idx].axhline(u_02, label=r"$u_2$", c="red")
    ax[idx].axhline(0.0, label=r"$u_1$", c="black")
    ax[idx].plot(x, sim_matrix[:, time])
    ax[idx].set_xlabel("Position")
    ax[idx].set_ylabel("Concentration")
    ax[idx].set_title(f"Time = {time*time_step:.0f}")
    ax[idx].grid()
    ax[idx].legend()

fig.suptitle(rf"$\xi_0 = $ {xi_0:.2f}, $u_0 = $ {u_02:.2f}")
fig.tight_layout()
fig.savefig("../report/figures/1b3.pdf")

t_min = int(40 / time_step)
t_max = int(300 / time_step)
delta_step = int(0.1 / time_step)

t_mid = int((t_max + t_min) / 2)

c = get_velocity_of_wave(sim_matrix, u_01, t_min, t_max, delta_step, time_step)
dudxi = euler_first_derivative(sim_matrix[:, t_mid], habitat_step)

fig, ax = plt.subplots(1, 2, figsize=(10, 5))
ax[0].plot(x, sim_matrix[:, t_mid])
ax[0].text(30, 1, f"c = {np.mean(c):.3f}")
ax[1].plot(sim_matrix[1:-1, t_mid], dudxi, label="derivative")
ax[1].scatter(sim_matrix[1, t_mid], dudxi[0], label="start point", color="red")
ax[1].scatter(sim_matrix[-2, t_mid], dudxi[-1], label="end point", color="black")
ax[1].legend()

ax[0].set_xlabel("x")
ax[0].set_ylabel("u(x)")
ax[1].set_xlabel("u(x)")
ax[1].set_ylabel("v(x)")

fig.tight_layout()
fig.savefig("../report/figures/1b3standingwave.pdf")

def plot4():
    rho = 0.5
    q = 8.0
    u_01 = 1 / 2 * (q - 1 + np.sqrt((q - 1) ** 2 - 4 * q * (1 / rho - 1)))
    u_02 = 1 / 2 * (q - 1 - np.sqrt((q - 1) ** 2 - 4 * q * (1 / rho - 1)))
    xi_0 = 50.0

```

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time_step = 0.1
sim_time = 1000
habitat_size = 100

sim_matrix = run_simulation(
    u_01, xi_0, rho, q, habitat_size, sim_time, time_step, second_init=True
)

plot_times = np.array([0, 50, 100, 150])
x = np.arange(1, habitat_size + 1, 1)
fig, ax = plt.subplots(2, 2, sharey=True)
ax = ax.flatten()

for idx, time in enumerate(plot_times):
    ax[idx].axhline(u_01, label=r"$u_3$", c="black")
    ax[idx].axhline(u_02, label=r"$u_2$", c="red")
    ax[idx].axhline(0.0, label=r"$u_1$", c="black")
    ax[idx].plot(x, sim_matrix[:, time])
    ax[idx].set_xlabel("Position")
    ax[idx].set_ylabel("Concentration")
    ax[idx].set_title(f"Time = {time*time_step:.0f}")
    ax[idx].grid()
    ax[idx].legend()

fig.suptitle(rf"$\xi_0 = $ {xi_0:.2f}, $u_0 = $ {u_01:.2f}")
fig.tight_layout()
fig.savefig("../report/figures/lc1.pdf")

def plot5():
    rho = 0.5
    q = 8.0
    u_01 = 1 / 2 * (q - 1 + np.sqrt((q - 1) ** 2 - 4 * q * (1 / rho - 1)))
    u_02 = 1 / 2 * (q - 1 - np.sqrt((q - 1) ** 2 - 4 * q * (1 / rho - 1)))
    xi_0 = 50.0
    time_step = 0.1
    sim_time = 1000
    habitat_size = 100

    sim_matrix = run_simulation(
        u_01 * 3, xi_0, rho, q, habitat_size, sim_time, time_step, second_init=True
    )

    plot_times = np.array([0, 100, 400, 2000])
    x = np.arange(1, habitat_size + 1, 1)
    fig, ax = plt.subplots(2, 2, sharey=True)
    ax = ax.flatten()

    for idx, time in enumerate(plot_times):
        ax[idx].axhline(u_01, label=r"$u_3$", c="black")
        ax[idx].axhline(u_02, label=r"$u_2$", c="red")
        ax[idx].axhline(0.0, label=r"$u_1$", c="black")
        ax[idx].plot(x, sim_matrix[:, time])
        ax[idx].set_xlabel("Position")

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        ax[idx].set_ylabel("Concentration")
        ax[idx].set_title(f"Time = {time*time_step:.0f}")
        ax[idx].grid()
        ax[idx].legend()

    fig.suptitle(rf"$\xi_0 = $ {xi_0:.2f}, $u_0 = $ {u_01 *3:.2f}")
    fig.tight_layout()
    fig.savefig("../report/figures/1c2.pdf")

def main():
    plot1()
    plot2()
    plot3()
    plot4()
    plot5()

if __name__ == "__main__":
    main()
```